

Lecture 7

- ① Pre-integration
- ② optimisation based VINS with IMU factor
- ③ Multi-IMU fusion

Pre-integration:

$$x_2 = f(x_1, n_1) \quad x_{k+1} = f_{int}(x_k, n_1) \quad \otimes \quad 1 \text{ measurement} : k \rightarrow k+1$$

$$x_j = f_{int}(x_k, n_1) \quad \otimes \quad \text{multiple measurements: } k \rightarrow j$$

$\otimes$: can be described as:

$$\begin{cases} {}^G I_j R = {}^G I_k R \cdot \Delta R \\ {}^G p_{2j} = {}^G p_{2k} + {}^G v_{2k} \Delta t_k + {}^G I_k R \Delta p + \frac{1}{2} {}^G g \Delta t_k^2 \\ {}^G v_{2j} = {}^G v_{2k} + {}^G I_k R \Delta v + {}^G g \Delta t_k \\ b_{gj} = b_{gk} + \Delta b_g \\ b_{aj} = b_{ak} + \Delta b_a \end{cases}$$

It is clear that ΔR , Δp and Δv are integration terms, then we can reorganize the measurements:

$$\begin{cases} \Delta R \triangleq \Delta R(b_g, n_1) \triangleq {}^G I_j R = \\ \Delta p \triangleq \Delta p(b_a, b_g, n_1) \triangleq \int_{t_k}^{t_j} \int_{t_k}^{t_s} {}^G I_s R {}^G a ds dt = \\ \Delta v \triangleq \Delta v(b_a, b_g, n_1) \triangleq \int_{t_k}^{t_j} {}^G I_s R {}^G v ds = \\ \Delta b_g \triangleq \Delta b_g(n_1) \triangleq \int_{t_k}^{t_j} n_{wg} dt = \\ \Delta b_a \triangleq \Delta b_a(n_1) \triangleq \int_{t_k}^{t_j} n_{wa} dt = \end{cases}$$

$$\begin{aligned} & {}^G I_k R^T {}^G I_j R \rightarrow h_R(x_k, x_j) \\ & {}^G I_k R^T ({}^G p_{2j} - {}^G p_{2k} - {}^G v_{2k} \Delta t_k - \frac{1}{2} {}^G g \Delta t_k^2) \rightarrow h_p(x_k, x_j) \\ & {}^G I_k R^T ({}^G v_{2j} - {}^G v_{2k} - {}^G g \Delta t_k) \rightarrow h_v(x_k, x_j) \\ & b_{gj} - b_{gk} \\ & b_{aj} - b_{ak} \end{aligned}$$

Left Hand side: the measurements from IMU (relates to IMU biases and noises)

Right hand side: The non-linear function containing the states x

$$\begin{cases} \Delta R(b_g, n_2) = h_R(x_k, x_j) \\ \Delta p(b_a, b_g, n_2) = h_p(x_k, x_j) \\ \Delta V(b_a, b_g, n_2) = h_V(x_k, x_j) \\ \Delta b_g(n_2) = b_{gj} - b_{gk} \\ \Delta b_a(n_2) = b_{aj} - b_{ak} \end{cases} \Rightarrow \text{The IMU measurements, no assumption yet.}$$

Give an estimate of $\hat{b}_g$ and $\hat{b}_a$, then:

$$\Delta R(b_g, n_2) \approx \Delta R(\hat{b}_g, 0) \cdot \text{Exp}(H_g^R \tilde{b}_g + H_n^R n_2)$$

$$\Delta R(\hat{b}_g, 0) = h_R(x_k, x_j) \cdot \text{Exp}(-H_g^R \tilde{b}_g - H_n^R n_2)$$

$$\text{similarly: } \Delta p(b_g, b_a, n_2) \approx \Delta p(\hat{b}_g, \hat{b}_a, 0) + H_g^p \tilde{b}_g + H_a^p \tilde{b}_a + H_n^p n_2$$

$$\Delta p(\hat{b}_g, \hat{b}_a, 0) = h_p(x_k, x_j) - H_g^p \tilde{b}_g - H_a^p \tilde{b}_a - H_n^p n_2$$

$$\Delta V(b_g, b_a, n_2) \approx \Delta V(\hat{b}_g, \hat{b}_a, 0) + H_g^v \tilde{b}_g + H_a^v \tilde{b}_a + H_n^v n_2$$

$$\Delta V(\hat{b}_g, \hat{b}_a, 0) = h_v(x_k, x_j) - H_g^v \tilde{b}_g - H_a^v \tilde{b}_a - H_n^v n_2$$

then for biases: we get:

$$\Delta b_g(n_2) = 0 + \int_{t_k}^{t_j} n_{ugd} dt = 0 + n_{ugd} \delta t$$

$$\delta t = t_j - t_k$$

n_{ugd} is the discrete noise

$$\Delta b_a(n_2) = 0 + \int_{t_k}^{t_j} n_{wad} dt = 0 + n_{wad} \delta t$$

n_{wad} is the discrete noise

Hence: the cleaner IMU measurement is:

$$\begin{cases} \Delta R(\hat{b}_g, 0) = h_R(x_k, x_j) \cdot \text{Exp}(-H_g^R \tilde{b}_g - H_n^R n_2) \\ \Delta p(\hat{b}_g, \hat{b}_a, 0) = h_p(x_k, x_j) - H_g^p \tilde{b}_g - H_a^p \tilde{b}_a - H_n^p n_2 \\ \Delta V(\hat{b}_g, \hat{b}_a, 0) = h_v(x_k, x_j) - H_g^v \tilde{b}_g - H_a^v \tilde{b}_a - H_n^v n_2 \end{cases}$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} b_{gj} - b_{gk} - n_{ugd} \delta t \\ b_{aj} - b_{ak} - n_{wad} \delta t \end{bmatrix}$$

$$\tilde{z}_1 = h(x_k, x_j, n_2)$$

$$n_{ij} \triangleq n_2 \sim \mathcal{N}(0, R_{ij})$$

$$\Rightarrow \tilde{z}_1 = \underbrace{h(\hat{x}_k, \hat{x}_j, 0)}_{\hat{\tilde{z}}_1} + H_k \tilde{x}_k + H_j \tilde{x}_j + G_1 n_{ij}$$

$$\Rightarrow \tilde{\tilde{z}}_1 = H_k \tilde{x}_k + H_j \tilde{x}_j + G_2 n_{ij}$$

why we would like this change? because we would to find this format:

$$\tilde{z}_1 = h(x_k, x_j, n_{kij}) \rightarrow \tilde{\tilde{z}}_1 = H_k \tilde{x}_k + H_j \tilde{x}_j + G_2 n_{kij}$$

then we can have $\| \tilde{z}_1 - h(x_k, x_j, 0) \|^2_{G_2 R_{ij} G_2^T} \Rightarrow \| \tilde{\tilde{z}}_1 - H_k \tilde{x}_k - H_j \tilde{x}_j \|^2_{G_2 n_{ij} G_2^T}$
a cost term:

This cost term can be used in BA/least squares with vision factor easily.

But there is a caveat for the above formulation:

- ① For the Least squares / MLE, we have multiple iterations
- ② For multi-iterations, $\hat{b}_g$ is changing, then: $\Delta p, \Delta R, \Delta v$ and G_1, n_{ij} will change
- ③ If so, For every iteration: we need to recompute: $\tilde{z}_1$ and G_1, n_{ij}
- ④ If we have high-frequency IMU and a lot of such IMU cost in the MLE, it will cause substantial computations.

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How to solve this.

$$z = h(x_A, x_B) + n$$

$$\hat{x}_A, \hat{x}_B^0$$

$$x_B = x_B^T + x_B^D$$

$$x_B = \hat{x}_B + \tilde{x}_B$$

$$= \hat{x}_B^0 + \hat{x}_B - \hat{x}_B^0 + \tilde{x}_B$$

$$= \hat{x}_B^0 + \Delta x_B + \tilde{x}_B$$

Then $\tilde{z} = H_A \tilde{x}_A + H_B^0 \tilde{x}_B^0 +$

$$= H_A \tilde{x}_A + H_B^0 (\Delta x_B + \tilde{x}_B) + n$$

$$\Rightarrow \tilde{z} = H_A \tilde{x}_A + H_B^0 \tilde{x}_B + \Delta x_B + n$$

$$\Rightarrow \tilde{z} + \boxed{\Delta x_B} = H_A \tilde{x}_A + H_B^0 \tilde{x}_B + n$$

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only this change
 small computation